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A Grice on Implicatures

The philosopher Paul Grice famously introduced the term *Implicature* in his William James Lectures of 1967 (Grice 1967), published in its most recent form in the chapter *Logic and Conversation* in Grice (1989).

- (25) Suppose that A and B are talking about a mutual friend, C, who is now working in a bank. A asks B how C is getting on in his job, and B replies, *Oh quite well, I think; he likes his colleagues, and he hasn't been to prison yet.* (Grice 1989, p. 24)

Grice goes on to say that B can be informally said to have implied, suggested or meant "...any one of such things as that C is the sort of person likely to yield to the temptation provided by his occupation, that C's colleagues are really very unpleasant and treacherous people, and so forth." (Grice 1989, p. 24) In order to enhance clarity of the discussion, he introduces the technical term *implicate*: B implicates – rather than says – that C is likely to succumb to the temptation of dishonesty, the proposition C IS LIKELY TO SUCCUMB TO THE TEMPTATION OF DISHONESTY is an *implicature* of (or rather, one of the implicatures of) B's utterance *he hasn't been to prison yet* in the example. *Saying* is in essence expressing the meaning of the conventional meaning of the sentence. So we can say that in the above example, B *says* that *C hasn't been to prison yet* and *implicates* that *C is likely to succumb to the temptation of dishonesty*, or that B's utterance conveys the meaning that *C is likely to succumb to the temptation of dishonesty* by way of *implicature*.

In other words, implicatures (more precisely *conversational implicatures* in Grice's terminology) are aspects of the communicator's intended meaning which can not be read off the conventional, linguistically encoded sentence meaning that the communicator has committed to conveying.²² Implicatures arise in the use of language in conversation, in communication. But how can audiences reliably comprehend such implicated meanings? Grice's answer was that conversation (communication) between rational agents is a cooperative endeavour, and what this amounts to is spelled out by a set of maxims of Quantity, Quality, Relation and Manner:

- (26) The Conversational Maxims of Grice (1989, pp. 26–27):

Cooperative Principle: Make your contribution such as is required, at the stage at which it occurs, by the accepted purpose or direction of the talk exchange in which you are engaged.

Quantity:

1. Make your contribution as informative as is required (for the current purposes of the exchange)
2. Do not make your contribution more informative than is required.

²²This reference to speaker's commitment is important because Grice claims that there are some implicatures which are connected to the conventional meaning of words such as *therefore*, which he calls *conventional implicatures*. However, these conventional implicatures are cancellable and do not contribute to the truth conditions of the utterance.

Quality: Try to make your contribution one that is true.

1. Do not say what you believe to be false.
2. Do not say that for which you lack adequate evidence.

Relation: Be relevant.

Manner:

1. Avoid obscurity of expression.
2. Avoid ambiguity.
3. Be brief (avoid unnecessary prolixity).
4. Be orderly.

A communicator conveying an implicature necessarily produces an utterance whose conventional (literal) meaning does not adhere to these maxims, at least not to all of them. However, the communicator can count on the audience's ability to see what the utterance might imply in the context at hand; if this implied import in turn satisfies the maxims, then the audience is justified to assume that the communicator intended to implicate this implied meaning, i.e. to convey this meaning as an implicature. Grice explains the pattern of inference for the comprehension of implicatures as follows:

- (27) A man who, by (in, when) saying (or making as if to say) that *p* has implicated that *q*, may be said to have conversationally implicated that *q*, provided that (1) he is to be presumed to be observing the conversational maxims, or at least the Cooperative Principle; (2) the supposition that he is aware that, or thinks that, *q* is required in order to make his saying or making as if to say *p* (or doing so in *those* terms) consistent with this presumption; and (3) the speaker thinks (and would expect the hearer to think that the speaker thinks) that it is within the competence of the hearer to work out, or grasp intuitively, that the supposition mentioned in (2) is required. (Grice 1989, p. 31)

Applying this pattern to his introductory example (25), Grice writes the following:

- (28) In a suitable setting A might reason as follows: "(1) B has apparently violated the maxim *e relevant'* and so may be regarded as having flouted one of the maxims conjoining perspicuity, yet I have no reason to suppose that he is option out from the operation of the Cooperative Principle; (2) given the circumstances, I can regard his irrelevance as only apparent if, and only if, I suppose him to think that C is potentially dishonest; (3) B knows that I am capable of working out step (2). So B implicates that C is potentially dishonest." (Grice 1989, p. 31)

There are two points to notice: first, it remains unexplained in (27) how the audience arrives at *q* given *p*. The reasoning process would certainly need to include some additional premises not derivable from *p*, and the question of how to choose these premises would need to be addressed (Wilson and Sperber 1991, p. 378). Second, notice that although this inference pattern is presented as a schema for a fully conscious and elaborate reasoning process, Grice does not claim that audiences necessarily apply this reasoning processes. In fact, he grants that audiences can often intuit implicatures subconsciously. However, he insists that hearers must be able to work out

the implicatures according to this schema. In other words, this working out schema for implicatures is not intended to provide a psychologically plausible inference pattern or procedure to identify implicatures. Rather, it serves as an analytical tool for theorists to describe the phenomenon.

B Code listing: Optimally relevant interpretive hypothesis in ProbLog – Int 1, Int 2

Interpretive hypothesis Int 1 is implemented as a ProbLog program as listed below. Int 2 differs from Int 1 by omitting lines 11–29.

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% Interpretive hypothesis 1: satisfying relevance expectations, optimal

% Modelling a web of easily accessible encyclopaedic assumptions
% as probabilistic graphs from beverages via categorizing nodes
% (e.g. tirednessBlockingDrink, partyDrink)
% to the superclass drink.
0.9::edge(coffee, tirednessBlockingDrink).
0.2::edge(coffee, performanceEnhancingDrink).
0.8::edge(coffee, partyDrink).
0.9::edge(water, partyDrink).
0.1::edge(water, performanceEnhancingDrink).
0.85::edge(tea, tirednessBlockingDrink).
0.2::edge(tea, performanceEnhancingDrink).
0.9::edge(tea, partyDrink).
0.4::edge(peppermintTea, tirednessBlockingDrink).
0.9::edge(peppermintTea, partyDrink).
0.8::edge(fruitTea, partyDrink).
0.01::edge(fruitTea, tirednessBlockingDrink).
0.01::edge(fruitTea, performanceEnhancingDrink).
0.8::edge(coke, tirednessBlockingDrink).
0.6::edge(coke, performanceEnhancingDrink).
0.8::edge(coke, partyDrink).
0.9::edge(redBull, performanceEnhancingDrink).
0.8::edge(redBull, tirednessBlockingDrink).
0.3::edge(redBull, partyDrink).
0.87::edge(proteinShake, muscleGrowingDrink).
0.87::edge(proteinShake, performanceEnhancingDrink).
0.1::edge(proteinShake, partyDrink).
edge(partyDrink, beverage).

% Definition of person
person(X) :- path(X, person).
% Mary is a person
edge(mary, person).

% The word energizingDrink could convey the concepts
% tirednessBlockingDrink; performanceEnhancingDrink; muscleGrowingDrink
.
energizingDrink(X) :- 0.9::path(X, tirednessBlockingDrink);
                    0.7::path(X, performanceEnhancingDrink);
                    0.7::path(X, muscleGrowingDrink).

% For categorization, we need performanceEnhancingDrink/1, etc.
performanceEnhancingDrink(X) :- path(X, performanceEnhancingDrink).

muscleGrowingDrink(X) :- path(X, muscleGrowingDrink).
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